

Giovanni Briglia - Ph.D. Student in Causal Reinforcement Learning

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🌐 Website

🌐 Scholar

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🌐 LinkedIn

Last Update: Sept. 2026

My research focuses on Causal Reinforcement Learning in both single- and multi-agent settings. I am studying the theoretical foundations of causality in sequential decision-making and developing methods to represent and integrate causal reasoning into the learning process, with the aim of improving generalization, sample efficiency, and interpretability.

Research Positions

- Sept 2026 - Nov 2026  **Ph.D. Visiting Researcher, University of Oxford - OXCAV**
Topics: Causality; Reinforcement Learning; Thinking Fast and Slow in AI.
Supervisors: Alessandro Abate and Francesco Fabiano.
- Nov 2024 - present  **Ph.D. Student, National PhD in AI**
Supervisors: Franco Zambonelli and Stefano Mariani.
- Jan 2025 - Feb 2025  **Ph.D. Researcher, Alan Turing Institute**
Selected for the Data Study Group of January and February. I participated in the C-DICE challenge, developing an optimization framework for repurposing energy assets to support the UK's Net-Zero 2050 goal.
- Nov 2023 - Nov 2024  **Research Fellow, Distributed and Pervasive Intelligence Group**
Topic: Causality-Driven RL.
Supervisors: Franco Zambonelli and Stefano Mariani.

Education

- Nov 2024 - present  **Ph.D. Computer Science, National PhD in AI**
Topic: Causal Reasoning in RL and MARL.
Courses: Markov Processes, Stochastic Processes, Game Theory, Distributed AI, Introduction to LLMs.
Supervisors: Franco Zambonelli and Stefano Mariani.
- Sep 2021 - Oct 2023  **M.Sc. Mechatronics, Robotics and Automation Engineering, University of Modena and Reggio Emilia**, Grade: 110 with honors/110.
Thesis title: Integrating Causality into Q-Learning for Adaptive Control in Dynamic Environments. *Supervisor:* Marco Lippi
- Oct 2022 - Mar 2023  **Erasmus+ exchange semester, Technische Universität München (TUM)**
Courses: Embedded Network Systems, Concept and Software Design for CPS, Experimental Vibration Analysis, Visual Data Analytics, Robotics.
- Sep 2018 - Oct 2021  **B.Sc. Mechatronics, Robotics and Automation Engineering, University of Modena and Reggio Emilia**, Grade: 96/110.
Thesis title: Artificial Intelligence applied to predictive maintenance.
Supervisor: Marco Lippi

Research Publications

Journal Articles

- 1 G. Briglia, F. Immovilli, M. Cocconcelli, and M. Lippi, "Bearing fault detection and recognition from supply currents with decision trees," *IEEE Access*, vol. 12, pp. 12 760-12 770, 2023.

Conference Proceedings

- 1 G. Briglia, F. Fabiano, and S. Mariani, "Scaling multi-agent epistemic planning through gnn-derived heuristics," in *The 25th International Conference on Autonomous Agents and Multi-Agent Systems*.

- 2 G. Briglia, "Causal learning and reasoning in multi-agent reinforcement learning," in *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems*, 2026, pp. 3975–3977.
- 3 G. Briglia, S. Mariani, and F. Zambonelli, "Causal models improve reinforcement learning for pervasive and robotic tasks," in *2026 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, IEEE, 2026, pp. 1–6.
- 4 S. Mariani, G. Briglia, A. Meyer-Vitali, M. Klusch, and F. Zambonelli, "Causal learning and reasoning in multi-agent systems: Goals, issues, and taxonomy," in *Proceedings of the 1st International Workshop on Causal Learning and Reasoning in Agents and Multiagent Systems, CLaRAMAS - Paphos, Cyprus, May 26, 2026*, In press, Springer, 2026.  URL: <https://openreview.net/pdf?id=FCr9YFHvwO>.
- 5 G. Briglia, S. Mariani, and F. Zambonelli, "Towards safe action policies in multi-robot systems with causal reinforcement learning," in *Workshop on Agents and Robots for reliable Engineered Autonomy*, Springer, 2025, pp. 51–71.
- 6 G. Briglia, M. Lippi, S. Mariani, and F. Zambonelli, "Improving reinforcement learning-based autonomous agents with causal models," in *International Conference on Principles and Practice of Multi-Agent Systems*, Springer, 2024, pp. 267–283.
- 7 G. Briglia, F. Immoilli, M. Cocconcelli, and M. Lippi, "Cross-load generalization of bearing fault recognition with decision trees," in *2023 7th International Conference on System Reliability and Safety (ICSRS)*, IEEE, 2023, pp. 400–406.

Pre-prints

- 1 G. Briglia, S. Mariani, and F. Zambonelli, *A roadmap towards improving multi-agent reinforcement learning with causal discovery and inference*, arXiv preprint arXiv:2503.17803, 2025.

Research Presentations

September 2026	 OXCAV Seminar: Given a 2 hours seminar about "When, Why, and How is Causal Reasoning Needed in Reinforcement Learning", hosted by prof. Alessandro Abate.
	 Erlangen AI Conference: Oral presentation and poster presentation about my vision of causality in multi-agent RL.
May 2026	 AAMAS 2026: Oral presentation of [3] at the doctoral consortium, about the vision of causality in multi-agent RL; and [2], describing our approach of using Graph Neural Networks in multi-agent epistemic planning.
March 2026	 IEEE PerCom 2026 - CoMoRe-AI workshop: Oral presentation of [4], describing our causal value function estimation procedure. slides link
October 2025	 ECAI 2025 - AREA workshop: Oral presentation of [6], demonstrating that independent causal learning is inadequate for tackling the intrinsic interdependencies of complex multi-agent systems. The work was recognized with the Best Paper Award. slides link
October 2024	 PRIMA 2024: Oral presentation of [7], showing my first iteration in integrating causal knowledge in Reinforcement Learning as an external action filter. slides link

Participation in Research Projects

Jan 2025 - Feb 2025	 Data Study Group - Alan Turing Institute: Collaborating with fellow PhD students on the challenge proposed by C-DICE, focused on developing a software -based, cost-effective roadmap to achieve net-zero decarbonization by 2050.
Feb 2024 - Nov 2024	 AGRARIAN: this project aims to create an advanced agricultural solution focused on utilizing both drones and rovers to gather images and data for analyzing vineyards, specifically targeting the detection of golden flavescence. <i>Supervisors:</i> Marco Lippi and Stefania Monica.

Participation in Research Projects (continued)

Mar 2021 - Aug 2023  **ProjectRED:** evolved from mechanical division member to R&D lead: designed electronic housing and ML modules for semi-adaptive suspension, managed a 15-member rover team, and pioneered autonomous rover localization and mapping with YOLOv5, homography, and SLAM.

Research Questions

- Representation**  *What form of representation (possibly learned and continuously updated in real time) best captures and infers transition dynamics and mechanisms both effectively and explainably? How can such representations be reused across environments, tasks, and policies?*
- Integration**  *To what extent, and through which mechanisms, can causal knowledge be integrated into RL so that it becomes an active component of policy optimization and value estimation? At which stage(s) of the RL process is this integration most effective?*
- Multi-Agent**  *How can causality-driven RL methods be extended to multi-agent settings, where cooperation and competition introduce additional layers of complexity and interdependence?*

Reviewer Activities

- 2026  CLaRAMAS, CWM
- 2025  ECAI
- 2024  IEEE TII

Participation in Summer Schools

- Mar 2025  **Learning Over Time** @National Phd on AI
- July 2024  **OxML Representation Learning and Generative AI** @University of Oxford
- May 2024  **OxML Fundamentals** @University of Oxford
- Sep 2022  **Advanced Course in AI** @AImageLab

Awards and Achievements

- October 2026  Best paper award at ECAI 2025 - AREA workshop
- July 2024  Ranked 1st out of 150 candidates in the selection for the *National Ph.D. Program in AI*
- June 2024  OxML 2024 partial scholarship
- November 2022  Best 20 recent graduates in Italy in the engineering area by AlmaLaurea

Certifications

- 2024  **Reinforcement Learning Specialization**, University of Alberta.
- 2023  **Deep Learning Specialization**, DeepLearning.AI.
-  **Game Theory**, Stanford Online.
- 2021  **English Language Certification: B2**, Trinity College of London.

Skills

Languages	Italian mother tongue and fluent in English
Coding	Python, C, C++, Matlab
Python Libraries	torch, scikit-learn, pandas, numpy, scipy, matplotlib
RL Libraries	stable-baselines3, cleanrl, minimalRL, RLLib, torchrl
Causality Libraries	causalnex, pgmpy, CausalML, EconML, gCastle, do-why, causal-learn, causallib
Distributed Systems	multi-threaded/processing design, parallel/distributed computing
Version Control	GitHub, Git-Lab, CI/CD
Operating Systems	Windows, Linux
Misc.	Academic research, teaching, training, consultation, Office, $\LaTeX$

I authorize the processing of personal data contained in my curriculum vitae on the basis of Legislative Decree 196/2003, coordinated with Legislative Decree 101/2018, and EU Regulation 2016/679.

